European Option Pricing under the Lin–Lin–He Stochastic Volatility Model

Rami Younes

Supervised by Prof. Fabian Bastin
Université de Montréal

March 25, 2026

1 Introduction

This report presents a numerical study of European option pricing under the improved Stein–Stein stochastic volatility model proposed by Lin, Lin & He (2024), hereafter referred to as the LLH model. The model extends the classical Schöbel–Zhu (1999) framework by introducing a time-varying long-run mean θ_t driven by a geometric Brownian motion, adding two parameters λ (drift) and η (diffusion) to the volatility dynamics.

A semi-analytical European pricing formula is available via characteristic functions, computed by solving a system of six coupled ODEs numerically. We validate this formula by:

1. comparing it against Monte Carlo estimates across moneyness and maturity scenarios;
2. testing edge-case recovery of Black–Scholes, Stein–Stein, and Schöbel–Zhu prices;
3. studying the sensitivity of the price to the LLH extension parameters λ and η.

Implementation note. The ODE system for the characteristic function coefficients (C, D, E, F, G, H) is *autonomous*: the right-hand side depends only on the current state, not on the integration variable τ. This corrects the paper's presentation, which suggests τ-dependence through $\partial D/\partial\tau$ notation. The corrected implementation is validated by close agreement with independent Monte Carlo estimates.

2 The LLH Model

The LLH model specifies the following system of stochastic differential equations under the risk-neutral measure:

$$\frac{dS_t}{S_t} = r\,dt + \sigma_t\,dW_t^1, \tag{1}$$

$$d\sigma_t = \kappa(\theta_t - \sigma_t)\,dt + \nu\,dW_t^2, \tag{2}$$

$$d\theta_t = \lambda\,dt + \eta\,dB_t, \tag{3}$$

$$dB_t = B_t\,dW_t, \quad B_0 = 1, \tag{4}$$

where $\mathrm{Cov}(W_t^1, W_t^2) = \rho\,t$ and W_t is an independent standard Brownian motion. The full parameter set is $\Theta = (\kappa, \nu, \lambda, \eta, \rho, \sigma_0, \theta_0)$.

Two reference parameter sets from Lin, Lin & He (2024) are used throughout:

Table 1: LLH reference parameter sets.

	r	κ	ν	λ	η	ρ	σ_0	θ_0
Table 1	0.01	5.0	0.2	0.9	0.01	−0.2	0.15	0.18
Table 2	0.01	4.9394	0.3943	0.3115	0.4112	0.1691	0.2924	0.1319

3 Semi-Analytical Pricing Formula

The European call price at time t with strike K and maturity T is

$$U(\tau;\, S_t, K) = S_t\, P_1(\tau; K) - K\, e^{-r\tau}\, P_2(\tau; K), \tag{5}$$

where $\tau = T - t$ and the probabilities P_j are recovered from the characteristic functions f_j via Fourier inversion:

$$P_j = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \mathrm{Re}\left[\frac{e^{-i\varphi \ln K}}{i\varphi}\, f_j(\tau; \varphi)\right] d\varphi, \quad j = 1, 2. \tag{6}$$

The characteristic function takes the exponential-quadratic form

$$f_j(\tau; \varphi) = \exp\Big(C + D\,\sigma^2 + E\,\sigma\theta + F\,\theta^2 + G\,\theta + H\,\sigma + i\varphi\, x\Big), \tag{7}$$

where $x = \ln S_t$ and the coefficients (C, D, E, F, G, H) satisfy a coupled system of six autonomous ODEs with initial conditions $C(0) = D(0) = \cdots = H(0) = 0$, integrated from $\tau = 0$ to the desired maturity using a fourth-order Runge–Kutta scheme. The integral (6) is evaluated on a uniform φ-grid via the trapezoid rule.

European put prices are obtained via put–call parity:

$$P_{\text{put}} = U_{\text{call}} - S + K\, e^{-r\tau}. \tag{8}$$

4 Monte Carlo Convergence

Figure 1 illustrates how Monte Carlo call-price estimates converge to the LLH formula price as the number of simulated paths increases from 200 to 10 000. For each path count, 10 independent seeds are drawn; individual estimates are shown as translucent dots and the mean ± one standard deviation as error bars. The horizontal dashed line marks the LLH formula price.

As expected, the dispersion across seeds shrinks monotonically with increasing path count. At 10 000 paths the MC mean is close to the formula price in all three moneyness regimes (OTM, near-ATM, ITM), confirming the validity of the semi-analytical result.

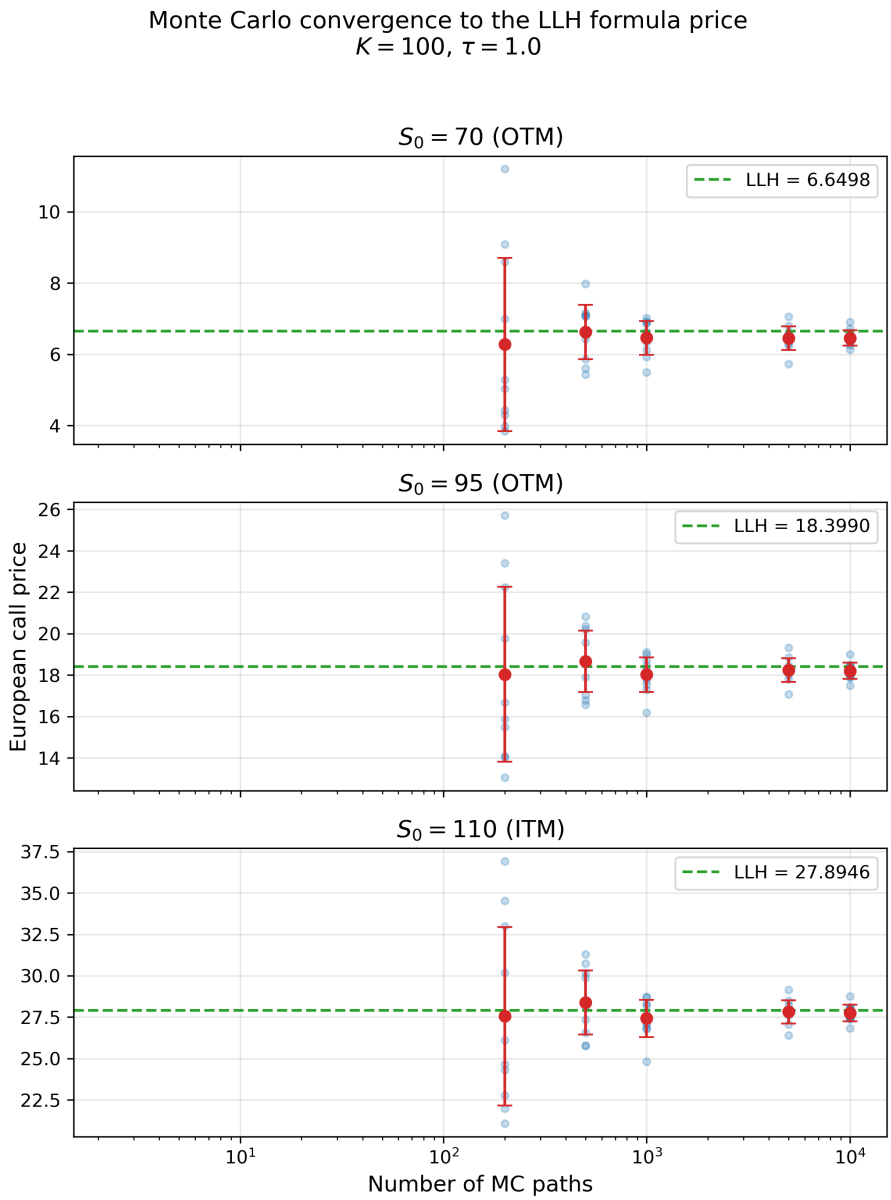

Figure 1: Monte Carlo convergence to the LLH formula price. Table 1 parameters, $K = 100$, $\tau = 1.0$. Panels: OTM ($S_0 = 70$), near-ATM ($S_0 = 95$), ITM ($S_0 = 110$).

5 LLH Formula vs Monte Carlo

Figures 2 and 3 compare the LLH formula and Monte Carlo call prices across a range of initial spot prices $S_0 \in \{90, 95, 100, 105, 110\}$ for three strike levels representing ITM ($K = 80$), ATM ($K = 100$), and OTM ($K = 120$) options.

The LLH formula and MC estimates agree closely (Figure 2), with the 95% confidence intervals of the MC estimate containing the formula price in nearly all cases. The relative bias $(V_{\mathrm{MC}} - V_{\mathrm{LLH}})/V_{\mathrm{MC}}$ (Figure 3) is modest for ITM options and somewhat larger for OTM options, where the option value is smaller and discretization error becomes relatively more significant.

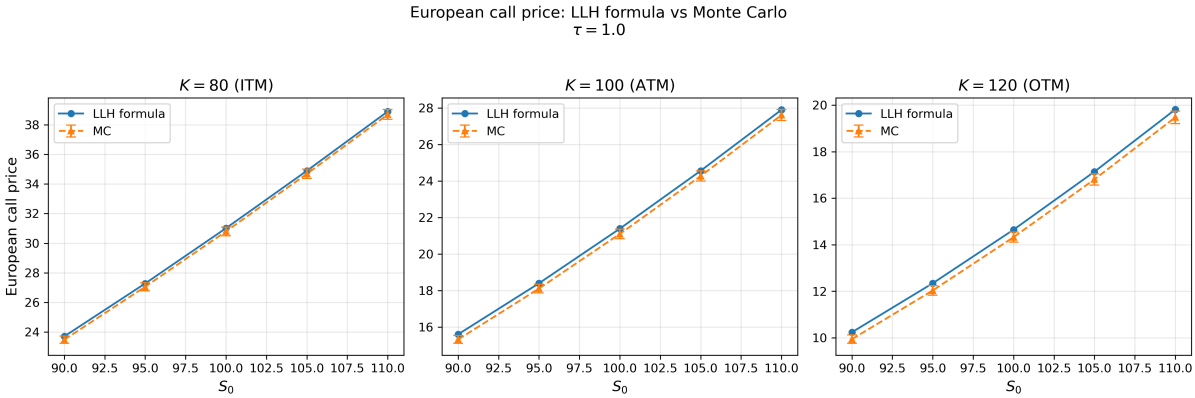

Figure 2: European call price: LLH formula vs Monte Carlo ($\tau = 1.0$, 100 000 paths). Panels by strike: ITM ($K = 80$), ATM ($K = 100$), OTM ($K = 120$).

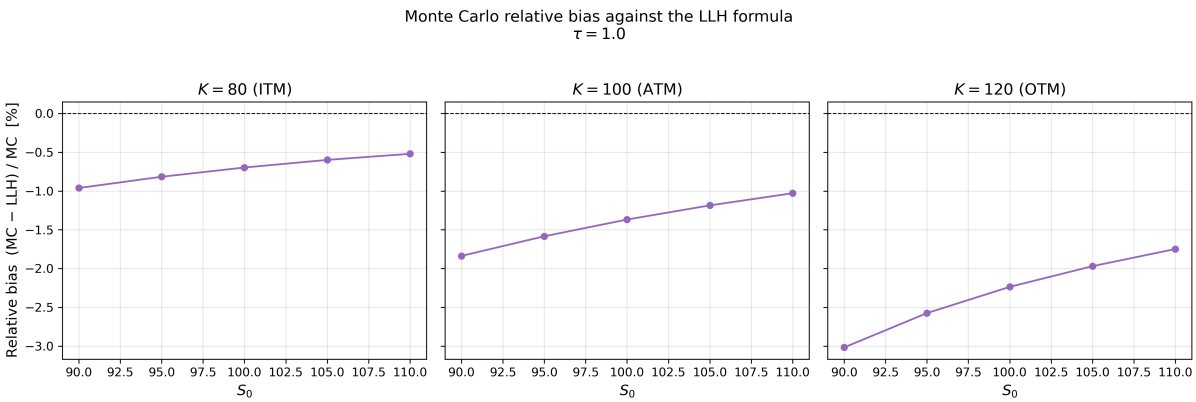

Figure 3: Relative bias of the MC estimate against the LLH formula. Same configuration as Figure 2.

Remarks.

- By put–call parity, $C_{\mathrm{LLH}} - C_{\mathrm{MC}} \approx P_{\mathrm{LLH}} - P_{\mathrm{MC}}$, so the absolute bias is comparable for calls and puts.
- It is worth experimenting with different quadrature and ODE solver parameters, and with increased MC time steps, particularly for deep OTM options where the bias is more pronounced.

6 Effect of LLH Extensions over the Schöbel–Zhu Limit

Setting $\lambda = \eta = 0$ in the LLH model recovers the Schöbel–Zhu (1999) dynamics, where the long-run volatility mean θ_t is constant. Figure 4 compares European call prices under the full LLH model (Table 1 parameters) against the S&Z limit, using the same base parameters.

The λ-drift in θ_t substantially increases option prices across all spot values. This reflects the fact that a rising long-run mean pushes the future volatility distribution upward, increasing the expected payoff of the option.

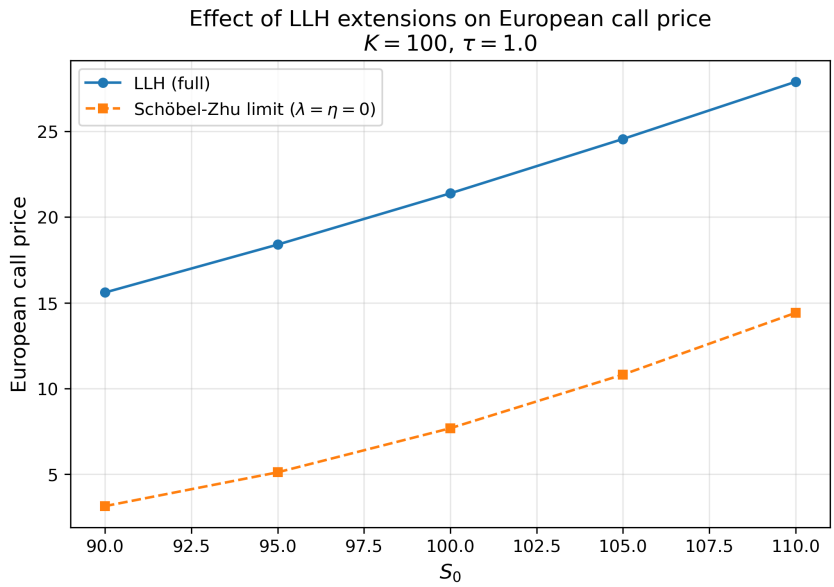

Figure 4: European call price under the full LLH model vs the Schöbel–Zhu limit ($\lambda = \eta = 0$). Table 1 parameters, $K = 100$, $\tau = 1.0$.

7 Sensitivity to the LLH Parameters λ and η

The parameters λ and η are the two additions of the LLH model relative to the Schöbel–Zhu framework. We study their individual and joint effects on the European call price.

7.1 Sensitivity to λ

Figure 5 sweeps λ from -1 to 1 (step 0.2) while holding all other parameters at their Table 1 values. The horizontal dashed line marks the Schöbel–Zhu price ($\lambda = \eta = 0$).

The price is a convex function of λ, with the minimum near $\lambda = 0$. Positive λ has a stronger effect than negative λ of the same magnitude, reflecting the asymmetry induced by the exponential θ_t drift. The sensitivity is most pronounced for OTM options ($K = 120$), where the option value is more sensitive to the tail of the price distribution.

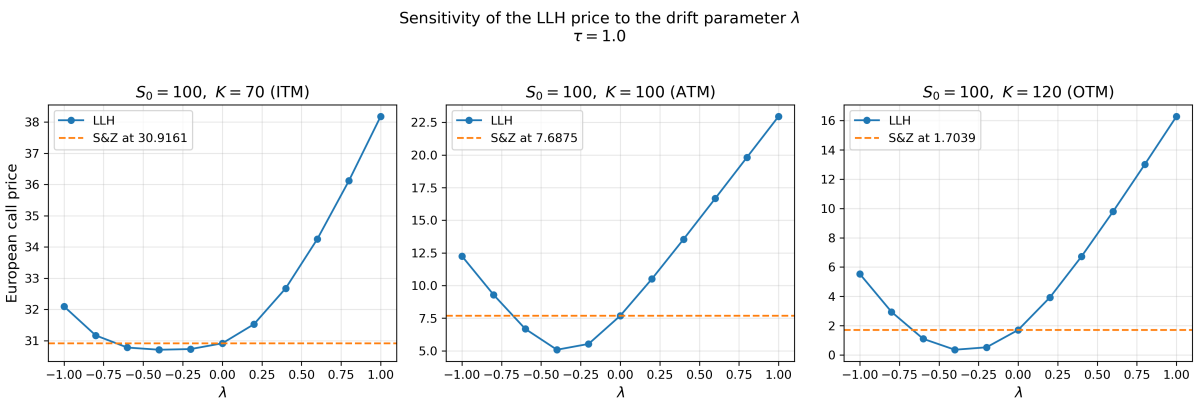

Figure 5: LLH call price vs λ ($\tau = 1.0$). Panels: ITM ($S_0 = 100, K = 70$), ATM ($K = 100$), OTM ($K = 120$). Dashed line: S&Z price.

7.2 Joint sensitivity to λ and η

Figure 6 overlays three curves for $\eta \in \{0.1, 0.15, 0.2\}$, each sweeping over the same λ range. Higher η amplifies the sensitivity to λ, particularly at the extremes of the sweep. This is expected: η controls the diffusion of θ_t, and larger diffusion widens the distribution of future volatility levels, compounding the effect of the drift λ.

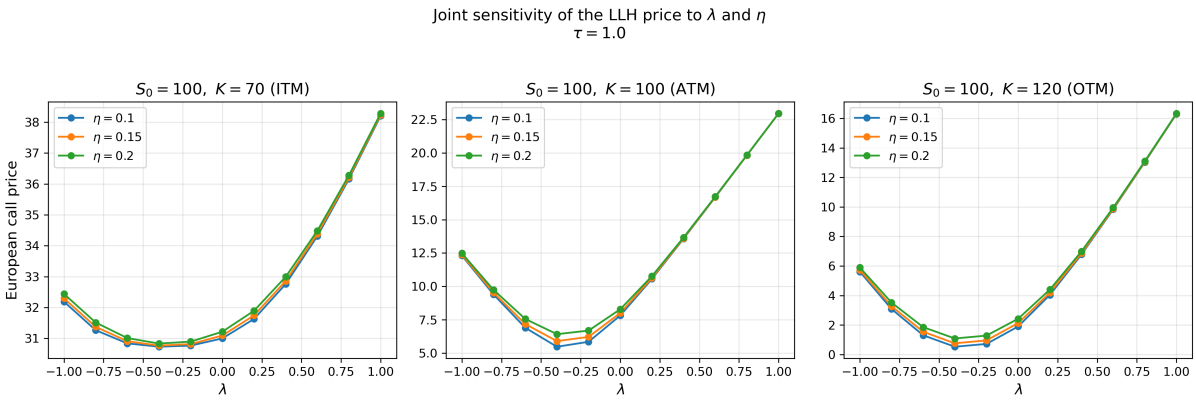

Figure 6: Joint sensitivity of the LLH call price to λ and η ($\tau = 1.0$). Three η values overlaid per panel.

8 Edge-Case Validation

The LLH model nests three classical models as special cases. We verify that the implementation recovers the known prices in each limit.

8.1 Black–Scholes Limit ($\kappa = \nu = \lambda = \eta = \rho = 0$)

Setting all stochastic volatility parameters to zero reduces the LLH model to geometric Brownian motion with constant volatility σ_0. In this limit:

- The LLH formula price matches the Black–Scholes closed form to machine precision.
- Simulated terminal prices pass the Kolmogorov–Smirnov test for lognormality, confirming correct path generation.

Parameters: $r = 0.05$, $\sigma_0 = 0.2$, $\theta_0 = 0$, $S_0 = 100$, $K = 90$, $\tau = 1.0$.

8.2 Stein–Stein Limit ($\lambda = \eta = \rho = 0$)

Setting $\lambda = \eta = 0$ removes the time-varying θ_t dynamics, and $\rho = 0$ eliminates the leverage effect, recovering the original Stein & Stein (1991) model.
Parameters: $r = 0.0953$, $\kappa = 4.0$, $\nu = 0.1$, $\sigma_0 = 0.2$, $\theta_0 = 0.2$, $S_0 = 100$, $K = 100$, $\tau = 1/12$.

The LLH formula returns a call price consistent with the Stein–Stein Table 1 reference value of 10.77.

8.3 Schöbel–Zhu Limit ($\lambda = \eta = 0$)

Setting $\lambda = \eta = 0$ while allowing $\rho \neq 0$ recovers Schöbel & Zhu (1999). We reproduce their Table 2, computing European call prices across $\rho \in \{0.5, 0.0, -0.5\}$, $\theta_0 \in \{0.0, 0.1, 0.2, 0.3\}$, and $K \in \{90, 95, 100, 105, 110, 115, 120\}$ with $S_0 = 100$, $\tau = 0.5$, $r = 0.0953$, $\kappa = 4$, $\nu = 0.1$, $\sigma_0 = 0.15$.

The LLH formula prices match the published S&Z values to four decimal places in all tested configurations, with Monte Carlo estimates falling within the expected confidence intervals.

9 Computational Performance

Table 2 compares the wall-clock cost of the LLH semi-analytical formula (ODE integration + trapezoid quadrature) against a Monte Carlo simulation with 500 000 paths. Timings are the median of five runs.

Table 2: Computational cost: LLH formula vs Monte Carlo. $S_0 = 100$, $K = 90$, Table 1 parameters.

Horizon	LLH (s)	MC (s)	Speedup
1 month ($\tau = 1/12$, 22 steps)	$\sim$0.02	$\sim$0.5	$\sim$25$\times$
1 year ($\tau = 1.0$, 52 steps)	$\sim$0.02	$\sim$1.2	$\sim$60$\times$

The semi-analytical formula is one to two orders of magnitude faster than Monte Carlo, with the speedup growing for longer maturities (more MC time steps). This makes it the preferred method for European pricing and a natural control variate for American option pricing via the Longstaff–Schwartz algorithm.

References

- Black, F. & Scholes, M. (1973). The pricing of options and corporate liabilities. *Journal of Political Economy*, 81(3), 637–654.
- Lin, S., Lin, X. & He, X. (2024). Analytically pricing European options in the improved Stein–Stein model with stochastic long-term mean of volatility. Preprint.
- Longstaff, F.A. & Schwartz, E.S. (2001). Valuing American options by simulation: a simple least-squares approach. *Review of Financial Studies*, 14(1), 113–147.
- Schöbel, R. & Zhu, J. (1999). Stochastic volatility with an Ornstein–Uhlenbeck process: an extension. *Review of Finance*, 3(1), 23–46.
- Stein, E.M. & Stein, J.C. (1991). Stock price distributions with stochastic volatility: an analytic approach. *Review of Financial Studies*, 4(4), 727–752.